

# SHADOW: LINUX HARDENING TOOL

## ABSTRACT

Linux systems are the backbone of modern IT infrastructure, powering servers, cloud platforms, enterprise networks, and cybersecurity environments. However, maintaining their security requires continuous monitoring, proper configuration, and regular hardening. Traditional security audits are often performed manually, making them time-consuming, error-prone, and difficult to manage consistently. To address these challenges, this project presents **Shadow**, a comprehensive terminal-based Linux system hardening tool that automates security assessment, vulnerability detection, risk analysis, and secure remediation while ensuring safe backup and rollback capabilities.

Shadow is developed using **Python 3.10+** and integrates with **systemd** to support both boot-time and on-demand security assessments without consuming persistent system resources. The tool utilizes **PyYAML** for configuration management, **Jinja2** for HTML report generation, **Requests** for HTTP communication, and standard Linux security components such as **PAM**, **iptables/UFW**, **OpenSSH**, and **systemd** services. It is designed to support popular Linux distributions including **Kali Linux**, **Ubuntu**, and **Debian**.

Unlike manual security audits, custom shell scripts, commercial security scanners, or limited open-source tools that mainly focus on vulnerability detection, Shadow provides an integrated solution that combines automated scanning, intelligent risk assessment, safe security hardening, detailed reporting, and complete rollback functionality. The tool follows a modular architecture consisting of **14 security categories**, including authentication, remote access, network security, file permissions, services, storage, monitoring, updates, kernel configuration, process management, audit logging, access control, scheduled tasks, and file integrity. Each module follows a standardized workflow for security checking, automated fixing, and restoration.

The execution process begins when a user selects an operation such as **--scan**, **--harden**, **--interactive**, **--restore**, or **--report**. The system then coordinates security modules to analyze the Linux environment, identify vulnerabilities, calculate an overall security risk score ranging from **0 to 100**, and classify the system into security levels from **LOW** to **CRITICAL**. Before applying any security modification, Shadow utilizes an atomic transaction engine with SHA-256 verified backups to ensure complete, module-level recovery if post-validation fails. After execution, the tool generates professional reports in **Terminal**, **JSON**, **HTML**, and **PDF** formats while maintaining comprehensive logs of all detected issues, applied fixes, and system changes. One of the major security features implemented in Shadow is an automated **three-attempt account lockout mechanism** using **PAM fail lock** with **deny=3** and **unlock\_time=600** seconds. This mechanism helps protect Linux systems against brute-force login attacks by temporarily locking user accounts after three consecutive failed authentication attempts.

The project consists of 89 source files with more than 5,000 lines of production-ready code, demonstrating a scalable and maintainable approach to Linux security automation. During testing, Shadow successfully completed full system security scanning and hardening operations in under two minutes. Furthermore, hardening results were independently cross-validated using Lynis 3.1.6 (the industry-standard Linux auditor), which confirmed all remediated areas as secure and verified a complete risk reduction from 79/100 (CRITICAL) to 0/100 (LOW) in a single automated pass.

Overall, Shadow provides a unified, production-ready, and open-source Linux security hardening solution (publicly available on GitHub) that simplifies system security management through intelligent automation. By combining comprehensive vulnerability assessment, automated remediation, risk-based analysis, professional reporting, and reliable rollback mechanisms, the tool significantly reduces administrative effort, improves security consistency, and enhances the overall protection of Linux-based systems.